

SALO KIIKALA, Finland

WMO: 027770

Lat: 60.464N

Lon: 23.650E

Elev: 117

StdP: 99.93

Time Zone: 2.00 (EUE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -20.9	(c) -17.7	(d) -23.2	(e) 0.5	(f) -20.7	(g) -19.8	(h) 0.7	(i) -17.4	(j) 8.9	(k) -3.6	(l) 8.2	(m) -4.6	(n) 2.1	(o) 30	(p) 0.614

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	27.0	18.1	25.2	17.4	23.4	16.4	20.0	24.7	18.7	23.1	17.5	21.6	4.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	13.4	22.0	17.0	12.3	20.3	15.9	11.5	19.0	57.7	24.7	53.4	23.2	49.6	21.5	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.5	6.8	6.1	-24.4	29.0	4.0	2.2	-27.3	30.6	-29.7	31.9	-31.9	33.1	-34.9	34.7
			-24.5	20.9	3.9	1.6	-27.3	22.1	-29.6	23.0	-31.8	23.9	-34.6	25.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.4	-5.5	-5.3	-2.1	3.8	10.2	14.1	17.2	15.7	11.2	5.1	1.1	-1.7	
	DBStd	8.94	6.06	5.90	4.72	3.42	4.34	3.51	3.09	2.97	3.14	3.74	4.26	4.99	
	HDD10.0	2341	481	429	376	188	51	6	0	1	23	157	267	362	
	HDD18.3	4778	739	663	634	436	254	135	61	92	215	411	517	621	
	CDD10.0	651	0	0	0	2	57	128	223	177	59	4	0	0	
	CDD18.3	44	0	0	0	0	2	7	25	9	0	0	0	0	
	CDH23.3	409	0	0	0	0	28	79	229	73	0	0	0	0	
	CDH26.7	64	0	0	0	0	3	7	44	10	0	0	0	0	
Wind	WSAvg	3.1	3.2	3.2	3.2	3.0	3.1	3.1	2.8	2.9	2.9	3.1	3.3	3.5	
Precipitation	PrecAvg	693	59	42	36	36	36	61	70	80	59	74	74	64	
	PrecMax	887	105	108	79	62	78	103	157	171	126	185	177	162	
	PrecMin	525	10	2	6	8	3	19	6	6	19	20	10	16	
	PrecStd	93	25	26	16	16	20	23	38	38	29	38	33	31	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.0	4.9	11.2	18.9	26.3	27.7	30.1	28.1	21.4	14.5	9.7	8.0	
		MCWB	4.8	2.7	5.3	10.5	15.2	17.2	19.2	19.2	16.2	12.0	9.0	7.4	
	2%	DB	3.7	3.1	7.7	16.0	23.5	25.6	27.8	25.2	19.2	12.7	8.6	6.2	
		MCWB	3.3	2.2	4.0	8.3	14.4	16.3	19.2	18.1	15.1	11.4	8.2	5.7	
	5%	DB	2.6	2.1	5.2	13.1	21.3	23.6	25.9	23.2	17.6	11.7	7.6	5.2	
		MCWB	2.4	1.6	2.7	7.3	13.3	15.7	18.3	16.8	14.1	10.8	7.1	4.6	
	10%	DB	1.6	1.2	3.3	10.6	18.7	21.3	24.0	21.4	16.2	10.7	6.6	4.1	
		MCWB	1.3	0.8	1.8	5.8	12.0	14.5	17.3	16.2	13.7	9.8	6.2	3.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.9	3.4	6.8	11.1	17.7	19.9	21.6	20.9	17.1	12.8	9.3	7.7
			MCDB	5.0	4.2	10.5	17.7	23.3	24.8	26.4	25.0	19.2	13.6	9.6	8.0
2%		WB	3.4	2.4	4.4	9.3	15.7	17.8	20.4	19.2	16.0	11.9	8.2	5.9	
		MCDB	3.6	3.0	6.7	14.7	20.4	23.2	25.2	23.0	18.5	12.4	8.5	6.1	
5%		WB	2.4	1.7	3.1	7.6	14.4	16.4	19.4	18.0	15.0	11.0	7.2	4.7	
		MCDB	2.7	2.1	4.6	11.9	19.6	21.5	24.2	21.6	16.8	11.6	7.5	5.0	
10%		WB	1.4	0.8	2.1	6.3	12.7	15.1	18.1	17.1	14.1	9.9	6.3	3.6	
		MCDB	1.5	1.2	3.2	9.9	18.1	20.1	22.9	19.9	15.7	10.5	6.6	4.1	
Mean Daily Temperature Range		5% DB	MDBR	4.8	5.3	7.2	9.2	10.7	9.7	9.5	8.9	7.7	5.9	4.0	4.1
			MCDBR	3.7	4.9	9.0	14.5	14.3	13.2	12.5	11.8	9.8	6.7	4.3	4.0
	MCWBR		3.8	4.3	5.9	8.2	6.9	6.0	5.3	5.6	6.0	5.2	4.2	3.9	
	5% WB	MCDBR	3.7	4.5	7.9	12.9	11.8	11.0	10.7	9.6	8.5	5.7	4.3	3.8	
		MCWBR	3.8	4.0	5.5	7.8	6.2	5.7	5.2	5.1	5.7	4.9	4.2	3.7	
Clear-Sky Solar Irradiance	taub	0.265	0.302	0.326	0.361	0.357	0.353	0.378	0.369	0.341	0.326	0.293	N/A		
	taud	2.124	2.197	2.275	2.352	2.422	2.456	2.406	2.425	2.498	2.411	2.213	N/A		
	Ebn at Noon	481	680	796	836	869	878	842	820	786	662	444	N/A		
	Edh at Noon	42	71	91	102	102	100	103	94	74	58	39	N/A		
All-Sky Solar Radiation	RadAvg	0.24	0.81	2.31	3.86	5.25	5.49	5.19	4.07	2.50	1.09	0.30	0.14		
	RadStd	0.04	0.17	0.28	0.39	0.52	0.45	0.51	0.43	0.32	0.18	0.06	0.02		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (59 neighbors)	+0.44	N/A	N/A	N/A	N/A	N/A	-135	N/A	N/A	N/A

Nomenclature: See separate page